The future of energy- manuscript

SLIDE 1: Welcome

(Karin)

Estimate time use: 1 minute

Slide 1

Hi! Good morning, everyone! Welcome to our workshop about the green transition and renewable energy!

My name is [insert name], and this is [insert name]. We are master students/summer students/researchers at the University of Oslo.

Our day will consist of mini-lectures, informational videos, group discussions and some interactive exercises. We encourage you to actively participate, share your own thoughts and ideas and ask questions.

We also want to highlight the fact that what we are learning today isn't just academic knowledge, its important knowledge that we can use to change the course of our future.

Let's get started!

SLIDE 2: Agenda

(Karin)

Estimate time use: 1 minute

We have divided this workshop into three parts. The first part is a knowledge section about the green transition. This part is quite information-heavy, we will talk about how much energy we use in Norway, how much we produce as well as different forms of renewable energy.

The second part of this workshop is about why it's not just a matter of building more renewable energy in Norway. We will discuss and debate different opinions about renewable energy together.

The third part of this workshop is about climate justice. Here you will learn about international climate negotiations and "participate" in one yourself.

SLIDE 3: Who are you?

(Karin)

Estimate time use: 5 minutes

But first, we want to know more about you guys!

Thats why we wanted to do a short ice breaker where you introduce yourselves and name one thing you like to do, for example reading, knitting, Fortnite, dancing, et cetera, et cetera.

We can start. My name is [insert name], and one thing I like to do is.... - and my name is [insert name] and I like ....

SLIDE 4: Questionnaire

(Karin)

Estimate time use: 10-15 minutes

As mentioned, we teaching this workshop today are researchers at the University of Oslo. This teaching is part of a research project called "The Future of Energy." In this research project, we want to find out what climate change and the green transition mean for future generations, the today's youth.

To answer this question, we would like to collect your thoughts and opinions. It is voluntary, but we would be very grateful if you chose to answer our questionnaire.

The answers you give will be anonymous. To ensure this, you will now receive a code from me, do not share it with anyone. You should enter this code as the answer to the question in the form that asks exactly which code you have received. It is also important that you remember which number you received till the last day of the workshop. I would recommend you take a picture of this code with your cell phones.

Do you have any questions? If not, you can access the questionnaire using the link in the presentaiton.

SLIDE 5: What do you already know?

(Paola)

Estimate time use: 5 minutes (had to delete this side because of time)

Now we are finally starting! As a small warm-up exercise, we would like to hear your thoughts on what the big, complex, and quite difficult concept "the green shift" means or what it makes you think of. Please discuss this with the person sitting next to you: "what do you think when you hear the phrase 'the green shift'?"

[2 min discussion]

Good to see that you are so engaged in the discussions! Now, we would like you to type in some of the keywords from your discussion in mentimeter from your pc's.

[Discuss briefly what the students wrote on Mentimeter].

SLIDE 6: Climate change

(Paola)

Estimate time use: 5 minutes

Before we discuss what, the green transition refers to, we need to take a step back. We must talk about why the term "green transition" has emerged as a concept, expression, or even change process.

The reason why we in Norway are discussing the green transition is, of course, climate change. But what is climate change? Climate change refers to changes in weather and temperature over time (usually decades), from the way it usually is. An example of climate change are places in Norway that used to get a lot of snow over a long period now get less to no snow.

But what causes climate change? People have different opinions about this, but scientists have determined that the main cause is the increase of greenhouse gases in the atmosphere, such as carbon dioxide and methane. These gases get trapped in the atmosphere and retain heat from the sun, leading to a warmer climate.

The consequences of climate change are numerous. We experience more extreme weather, such as heatwaves, droughts, and severe storms. The ice caps are melting, sea levels are rising, and many animal species face the risk of extinction.

But why is this important to us? Because climate change affects everything from food production to our general health and safety. It is a global issue that requires action from everyone, including us here in Norway.

National Geographic has a video summarizing what climate change is all about, which you will see now.

Ask in plenary: Was this something you already knew, or did you learn something new?

SLIDE 7: The green transition

(Paola)

Estimate time use: 2 minutes

To stop climate change, society must operate in a more environmentally friendly way than it does today. We call this societal change towards a more environmentally friendly and sustainable course, "the green transition".

More concretely the green transition is about switching to more environmentally friendly products and services that do not have negative consequences for our environment and climate. Our consumption should not come at the expense of our nature.

Production and use of energy is a crucial component of the green shift. We must reduce our consumption of non-renewable energy sources to minimize the emission of greenhouse gasses. Non-renewable energy comes from energy sources that are stored in nature and, within a human time frame, do not renew themselves. Fossil fuels like oil, coal and natural gas are examples of non-renewable energy sources.

We must start using more renewable energy and less fossil fuel. But what exactly are we talking about when we say renewable energy? Let's explore that.

SLIDE 8: What is renewable energy?

(Karin)

Estimate time use: 5 minutes

Markus Bailey and Statkraft made a video where they explain what renewable energy truly is. While you watch the video, note down:

- What is renewable energy?
- Which renewable energy sources is Markus Bailey talking about?

Ask in plenary: What did you answer to the question what is renewable energy? Which renewable energy sources did Markus Bailey talk about in the video?

SLIDE 9: Energy use Norway

(Karin)

Estimate time use: 2 minutes

Now that we know the difference between renewable and non-renewable energy, let's look at how we use energy here in Norway.

The total annual energy consumption in Norway (in 2022) on the mainland and in the sea surrounding Norway/the continental shelf (largely from platforms in the North Sea) is over 300 TWh. Terawatt hours are a unit of measurement for energy. To give you an idea of how much that is, the entire municipality of Drammen (approximately 140,000 inhabitants) uses about 1 TWh in a year.

The graph you see here, shows the energy consumption in Norway the last 30 years. If we exclude the continental shelf from the calculation, Norwegian energy consumption was at 219 TWh. This usage includes all types of energy: electricity, gas, coal, oil, bioenergy, and district heating.

As you can also see from this graph, energy consumption has steadily increased. The use of gas and fossil fuels has decreased slightly in recent years. Also, electrification means higher efficiency, thus the more you electrictrify the less energy you use.

SLIDE 10: Energy use per sector

(Karin)

Estimate time use: 4 minutes

Ask in plenary: So, what do you think is all the energy we use in Norway is going to?

Here is the answer, the graph shows the amount of energy the different sectors use (2022). As we can see industry and mining use the most energy, followed by transportation, households, services, agriculture and fishing.

The graph also shows us what type of energy the different sectors use. The blue bars represent electricity, and the turquoise bars represent fossil fuels.

We can see the sector that uses the most fossil fuel is the transportation sector. While the industry-, household- and service sectors mainly rely on electricity.

But we can expect this distribution to change in the next couple of years. We can see this already happening with the increased focus on the electrification of transport. Eventually, this column here (transport) will also turn blue. In industry there is an ongoing switch towards using more electricity. Additionally, increased electricity usage is expected in new industries and data centers.

In other words, Norway must ramp up the production of renewable energy if we are to keep up with our needs.

SLIDE 11: Norway loves electricity

(Karin)

Estimate time use: 1 minute

Norway loves electricity, also known as electric power or simply power. Electricity is the most used form of energy in Norway. Approximately half, about 130 TWh, of the energy we use in Norway is electricity.

We use electricity for almost everything- from heating our homes to production of things like aluminum. And the good news is that:

A big proportion of this energy comes from large resources of the cheap renewable energy source, hydropower, which produces and emits less greenhouse gases!

Statnett expects an increase in electricity demand due to the electrification of e.g. transport and new industries.

SLIDE 12: What affects energy use? I

(Paola**)**

Estimate time use: 2 minutes

So, what influences how much energy we use in Norway? Norway's energy consumption is influenced by several factors. The first one being the population.

When we electrify our energy systems it becomes more efficient: e.g. replacing technologies or processes that use fossil fuels, like internal combustion engines and gas boilers, with electrically-powered equivalents, such as electric vehicles or heat pumps.

The population density also affects energy consumption. Urbanization leads to more people living in apartments, resulting in less living space per person which then requires less energy for heating. Centralization also makes is easier for people to use public transportation, cycle or walk to a destination due to shorter traveling distances, which means less energy used.

SLIDE 13: What affects energy use? II

(Paola**)**

Estimate time use: 4 minutes

Norway's energy consumption, or energy demand is also influenced by economic growth and new technology. Economic growth happens when production of goods and services increases in a society.

Ask in plenary: How do you think economic growth and new technology affects our energy consumption? Do you think technological advancements will lead to us consuming more or less energy?

Those are some very good suggestions, and in a way, you are all correct. Let's talk about why that's the case.

SLIDE 14: What affects energy use? III

(Paola**)**

Estimate time use: 2 minutes

The impact of economic growth and new technology on energy consumption is connected and can contribute to both increasing and decreasing our energy consumption.

Firstly, economic growth leads to increased demand for goods and services, thereby increasing the need for energy, both in the production and transportation of goods and services.

New technology can increase our energy consumption through the introduction of new machinery and appliances that require energy. Technological advancements also lead to increased demand for and production of goods and services which results in increased energy consumption.

New technology can also lead to reduced demand for goods and services, thereby reducing our energy consumption. It can also help dampen the growth in energy consumption by enabling us to use energy more efficiently. This more efficient use of energy means using less energy for the same job or activity as before. Both energy efficiency and reduced consumption (or reuse) are important parts of the green transition.

This is quite complex, so we'll try to explain this by using two examples.

SLIDE 15: New technology, more og less energy use? I

(Paola)

Estimate time use: 2 minutes

Electric cars, or EV's, are a good example for showcasing the complexity of new technology on energy consumption.

As you may already know EV's use electricity to power the motor instead of gasoline or diesel. The EV has changed the way we fuel our cars from using fossil fuels to electricity. Electric cars use energy 30-40% more efficiently than fossil fuel cars.

On the other hand, the amount of EV's in Norway has increased significantly in recent years – the demand has been high, leading to economic growth. This graph shows how many passenger cars we have in Norway. The blue line represents the number of EV's, we can see that in 2022, 4 out of every 5 new cars were EVs. The amount of electricity that goes into charging an EV for a 100km trip is the same amount that we use when we take a 35-minute shower. We also know that some of these cars were used as "second cars", which ultimately contributed to more production, economic growth and therefore increased energy consumption.

SLIDE 16: New technology, more or less energy use? II

(Paola)

Estimate time use: 2 minutes

New technology can also reduce our consumption of goods and services, resulting in less energy use.

Digital meeting platforms can for example lead to less air travel which reduces energy consumption.

Various apps like Bilkollektivet and Hyre have made it possible to share cars with others. Car sharing means that people who don't know each other have access to the same car. The car is usually owned by a company which rents it out it, and you can use it when you need to. Car sharing can reduce energy usage by reducing the number of cars produced. This way of "owning" a car replaces up to 10–15 private cars. Additionally, those who use car sharing drive 30 percent less.

One final example of an app that can help reduce our energy usage is Tise. Have you heard of it? This app allows you to buy used clothes, thereby reducing the production of new clothes and lowering energy consumption.

SLIDE 17: Break
(Paola)

Estimate time use: 5 minutes

Great job everybody! We're going to have a 5-minute break before we move on to our next topic, Norway's energy production.

SLIDE 18: Norway's Energy Production

(Karin)

Estimate time use: 4 minutes

So far today, we've talked a lot about the energy we use in Norway. Now, let's discuss the energy that is produced in Norway.

Pose question to the audience: What type of energy do you think is produced the most in Norway?

You guessed [correctly/almost correctly]! Here you see the answer. Was this surprising?

As you can see from the dark green and light green areas in the figure, Norwegian energy production primarily consists of oil and natural gas. The blue area represents the electricity produced in Norway, while the other smaller areas are coal, biofuels, and waste.

However, most of Norway's oil and gas are sold and used in Europe. The electricity is what is used in Norway. That's why earlier today we could say that Norway loves electricity.

SLIDE 19 and 20: Norway's electricity production
(Karin)
Estimate time use: 2 minutes

Since the green transition is critical, let's take a closer look at the electricity produced in Norway.

In 2023, Norway produced 154 TWh of electricity. As you may recall from earlier, this is more than we used We sold our electricity surplus (overskudd), and a bit more, to the United Kingdom, Denmark, Germany, Sweden, the Netherlands, and Finland. In 2023, we sold around 31 TWh of electricity, which is nearly 23% of Norway's total consumption. Why do we do this?

1. To ensure Norway has stable access to electricity.
2. To make a profit.
3. It's much easier to reduce emissions in the rest of the world that doesn't have the same access to renewable energy as we do.

In the graph being shown now, we can see how much electricity Norway has produced over the years. As you can see, we have produced significantly more electricity per year in the last 60 years, now producing over three times as much power as in 1965.

We also see that most of the electricity production in Norway comes from hydropower, while wind power has been increasing over the last 10 years.

SLIDE 21: Global Electricity Production
(Karin)
Estimate time use: 1 minute

When we compare Norway to the rest of the world, we see that most of the world's electricity is produced by burning fossil fuels (which emit a lot of CO_2 into the atmosphere), and that production has only been increasing.

The positive aspect of this picture is that electricity from renewable energy sources has also increased, perhaps relatively more than fossil fuels.

SLIDE 22: How is Electricity Transported?
(Shahazad)
Estimate time use: 2 minutes

So, how is all this electricity that's produced transported?

From a power plant that generates electricity, the electricity is transported through high-voltage power (or transmission) lines to our homes or others that use electricity.

There are two types of transmission lines: overhead and underground transmission lines. You've surely seen these around your neighborhood. The image in the middle shows an overhead transmission line; you can think of these as highways for electricity. Underground transmission lines can be thought of as long tunnels for electricity. The advantage of the underground lines is that we can't see them, but they are very expensive to install. In fact, they are 20 times more expensive than those placed above ground.

SLIDE 23: Hydropower
(Shahazad)
Estimate time use: 2 minutes

Now, let's take a deeper dive into the different sources of renewable energy used to generate electricity in Norway.

In a hydropower plant electricity is generated by turbines converting the movement of flowing water. In total, we have a whopping 1,769 hydropower plants. The beauty of hydropower is that it can be stored and used when needed.

The installed hydropower in Norway is estimated at 137.3 TWh. But can it go even higher?
The hydropower plants we have in Norway today can be upgraded to produce more electricity. Upgrading means building new ones, expanding, or refurbishing existing ones.

Excluding protected areas, Norwegian authorities estimate that the potential for upgrading is 7-8 TWh. For comparison, Norway generated 9.9 TWh from wind power in 2020.

When this potential is calculated, costs and environmental impacts are not considered, which can be significant when upgrading hydropower.

SLIDE 24: Onshore Wind Power
(Shahazad)
Estimate time use: 2 minutes

To increase electricity production, we must therefore look for alternatives to hydropower. One alternative is wind power. Here, we can distinguish between onshore (located on land) and offshore (located in water) wind power.

A wind farm consists of several wind turbines or windmills. These turbines generate electricity by converting the movement of the wind. In other words, electricity is produced only when the wind blows. Today, there are 65 wind farms in Norway with 1,392 wind turbines (2023, NVE).

How much power does a typical wind turbine produce? This depends on when the turbines were built since newer turbines are more efficient than older ones, but if the turbines were built in 2019, approximately 14 GWh. This is equivalent to the consumption of about 700 households.

SLIDE 25: Where in Norway are the Wind Farms Located?
(Shahazad)
Estimate time use: 4 minutes

Since wind power is "up and coming," let's look at where in Norway they are located.

Pose question to the audience: Where in Norway do you think wind farms are located?

Well, you guessed [correctly/almost correctly]! Here you can see the answer. Was this surprising?

As you can see in this figure, there are few wind farms located here at Østlandet, but in Trøndelag - as you may have heard in the news - you'll find most of our wind turbines.

Pose question to the audience: Why do you think the wind farms are located exactly here?

SLIDE 26: Where in Norway are the Wind Farms Located? II

(Shahazad)
Estimate time use: 1 minute (slide deleted because of time constraint)

On the map of Norway, the location of wind farms looks like this. The dark green circles you see represent developed wind farms, while the light green circles are under development.

SLIDE 27: Height of Wind Turbines I
(Shahazad)
Estimate time use: 1 minute

One of the discussions surrounding wind turbines is that they are quite visible. Larger wind turbines produce more electricity. Here is a picture from the Haram wind farm, taken nearly 3.5 km from the nearest turbine. These wind turbines are 150 meters high, but some are even 180 meters.

SLIDE 28: Height of Wind Turbines II
(Shahazad)
Estimate time use: 1 minute (slide deleted because of time constraint)

As we come even closer, such as 1.5 km here at the Frøya wind farm, the wind turbines look like this. These are somewhat taller than those on the previous slide, at 180 meters.

SLIDE 29: Height of Wind Turbines III
(Shahazad)
Estimate time use: 3 minutes

If we stand just a few meters away, a wind turbine at 150 meters at the Raudfjell wind farm looks like this.

Pose question to the audience: What do you think when you see these pictures? Was this larger or smaller than you had imagined?

SLIDE 30: Offshore Wind Power
(Shahazad)
Estimate time use: 2 minutes

An alternative to placing wind turbines on land is to place them at sea. This is widely discussed in Norway these days. In fact, Norway has recently granted the first permit to build an offshore wind farm in the North Sea.

Offshore wind turbines generate electricity in the same way as onshore wind turbines. Depending on the technology, they have the same capacity to produce energy.

The transportation of electricity is naturally different. Here, cables will be laid beneath the seabed so that the electricity can be transported to land and connected to the cables that further transport the electricity to consumers.

As you can see in the bottom picture, there are several ways wind turbines can be placed at sea. They can either be fixed and drilled into the seabed with a pole, or they can float. They are not completely floating as they are fixed in some way to connect to the cables that go to the mainland.

SLIDE 31: Wind Power in Norway, Sweden, and Denmark
(Shahazad)
Estimate time use: 1 minute

How much wind power (both onshore and offshore) do we produce in Norway compared to our neighbors? Despite Norway's long coastline, Sweden produces more wind power than Norway as of 2022. Denmark also produces more, and they are significantly smaller in size than Norway. Overall, in the Nordic region, Norway is behind in wind power production.

SLIDE 32: Solar Power
(Shahazad)
Estimate time use: 1 minute

Another alternative to increase electricity production from renewable energy sources is solar power. A solar power plant generates electricity by converting energy from the sun using solar panels.

Solar panels can be installed on any flat surface, from house or cabin roofs to directly on the ground. Like most renewable energy technologies, they require space, and a solar panel rarely stands alone. Like wind turbines, solar panels only produce energy if there is sunlight available.

SLIDE 33: Some Solar Installations and Solar Power Plants
(Shahazad)
Estimate time use: 1 minute

A solar panel (2 sqm) on a roof in Norway can produce between 650 - 1000 kWh/kWp per year. For a single-family home with 20,000 kWh in annual electricity consumption, this means that a solar panel system consisting of 20 panels could produce enough electricity to cover 25 percent of the home's electricity consumption.

The solar power plant you see in the picture can produce enough energy in a year for almost 4 Norwegian homes. In Norway, we have few large solar power plants, but this is continuously developing. However, we have many standalone solar installations, a total of 7691 (2022, NVE.no).

SLIDE 34: What do you think will be the consequences of the green shift in your municipality?
(Karin)
Estimate time use: 3 minutes

Wrapping up this day, we have an opinion poll. There are no right or wrong answers, but we would like to hear your opinion. Based on what we have discussed about the green transition, which involves changing to a more environmentally friendly society: What do you think will be the consequences of the green shift in your municipality?

Go to menti.com and give your answer!

SLIDE 35: Thank you for today! / Break!

(Karin)

Estimate time use: 1 minute

We hope you have learned something new today and that you have found this session exciting! We certainly had a great time here with you. Thank you for having us.

Today, you have received a lot of information and facts; next time, we will discuss challenges of wind turbines, and you will get to participate in a debate - we're looking forward to it.